SustAInable

A Supervised Learning Tool for Neighborhood-Level Heat Illness Risk Prediction

AI Memo — CTI 2026 Cohort — Equitech Futures Civic Tech Institute

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Unit of Analysis: ZIP Code Tabulation Area (ZCTA)

Domain: Climate Equity / Public Health

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Status: Design Memo — Seeking Development Partners

Executive Summary

Every year, extreme heat kills more Americans than hurricanes, tornadoes, and floods combined. The official US count for 2023 was 2,415 heat-related deaths. The estimated real count is closer to 11,000. The gap between those two numbers is a policy failure — and a data problem.

Two tools exist to help cities respond. NOAA HeatRisk tells emergency managers when a dangerous heat event is coming. The CDC Heat and Health Index identifies which neighborhoods are structurally vulnerable to heat. What neither tool answers is the question that actually matters five days before a heat event: given this specific forecast, which ZIP codes will see elevated emergency room visits and heat-related illness this time?

SustAInable answers that question.

SustAInable is a supervised machine learning model that assigns every US ZIP Code Tabulation Area (ZCTA) a probability score for elevated heat illness during an approaching extreme heat event. It integrates real weather forecasts with neighborhood-level vulnerability data — income, housing stock, disability prevalence, land surface temperature, chronic disease burden, and prior heat illness rates — to give cities, community organizations, and public health departments an actionable targeting tool before the crisis, not after.

Heat does not kill randomly. It follows patterns we can already see in our data. SustAInable makes those patterns actionable.

Motivation: Heat Kills — and It Does Not Kill Randomly

In July 1995, a heat wave descended on Chicago. Over five days, 739 people died. The bodies were not distributed across the city evenly. They concentrated in neighborhoods where poverty, housing stock, racial segregation, social isolation, and crumbling urban infrastructure combined into a lethal profile. Klinenberg's 2002 sociological analysis showed that two demographically similar Chicago neighborhoods had dramatically different death rates — separated not by temperature, but by whether people had working air conditioning, whether they trusted their neighbors, and whether there was commercial infrastructure that pulled them outside during the day. (Klinenberg, 2002, Heat Wave: A Social Autopsy of Disaster in Chicago.)

Nothing about that pattern has changed. What has changed is the frequency and severity of the events. Heat-related deaths in the US have risen 117% since 1999. Climate models project that extreme heat days will triple in frequency for US cities by mid-century. The cities that will suffer most are the ones already carrying the highest structural vulnerability.

I know this directly. I am a Disabled Philadelphian. I have Congenital Central Hypoventilation Syndrome (CCHS), a condition affecting the autonomic nervous system — the system that regulates breathing, heart rate, and critically, temperature. My wife Ariele has acquired disabilities including chronic illness with autonomic dysfunction. Heat is not an inconvenience for us. It is a physiological threat that our bodies cannot adequately self-regulate against.

We are not alone. Research published in the International Journal of Environmental Research and Public Health found that Disabled people face a 5x relative risk of heat illness compared to non-disabled people during extreme heat events. Elderly individuals, people with chronic cardiovascular and respiratory conditions, and residents of low-income housing without air conditioning face similar compounded risk. These communities are consistently the ones most likely to be missed by reactive emergency response.

SustAInable exists because Disabled people, elderly residents, and low-income communities deserve to be in the data model before the heat wave arrives — not as an afterthought in the after-action report.

The Existing Tool Gap — and What SustAInable Fills

Two federal tools do important work in this space:

- NOAA HeatRisk: A daily forecast tool that assigns a 0–4 danger level to US geographies based on expected temperature and humidity. Excellent at telling you when. Does not tell you where illness will be concentrated.
- CDC Heat and Health Index (HHI): A ZIP-code-level dataset of structural heat vulnerability factors — poverty rates, age distribution, housing age, green space. Excellent at telling you which neighborhoods are structurally vulnerable in the long run. Does not integrate real-time forecast data to predict illness during a specific upcoming event.

SustAInable bridges these tools. It takes the structural vulnerability context of the CDC HHI, integrates it with real weather forecast data from NOAA, and adds prior heat illness records at the ZIP-code level to produce event-specific predictions: during this heat event, these are the ZIP codes most likely to see elevated emergency room visits.

That distinction matters operationally. A city has finite cooling buses, wellness check teams, and emergency supplies. On day one of a heat emergency, a deployment decision cannot be made from a map of generalized structural vulnerability alone. It needs a ranked list: where do we send resources today? SustAInable produces that list.

Model Design

Label (Y): What the Model Predicts

The label is binary at the ZCTA level, for each heat event:

Label	Definition
1 (Elevated)	The ZCTA experienced elevated heat illness during the event: ≥ 2x baseline ED visits per 10,000 residents for heat-related illness (ICD-10 codes T67.x) during a NOAA HeatRisk Level 3+ event lasting ≥ 2 consecutive days.
0 (Not Elevated)	The ZCTA did not experience elevated heat illness during the event by the above threshold.

Label construction note: CDC HHI data is a 3-year aggregate, not event-level. To construct event-level labels, we supplement with CDC WONDER cause-of-death data and state-level syndromic surveillance (NSSP) emergency department data where available, matched to NOAA-declared heat events by geography and date. This is the most significant data engineering challenge in the project and is addressed directly in the risks section.

Inputs (X): What the Model Learns From

Features are organized into three categories:

Category	Features	Source
Structural Vulnerability	Poverty rate, % renters, % housing pre-1980 (poor insulation), disability prevalence, elderly population %, % without AC (ACS-estimated), green space coverage	ACS 5-Year Estimates; CDC Heat and Health Index; NLCD 2021
Health Burden	Cardiovascular disease prevalence, COPD prevalence, obesity rate, diabetes rate, prior heat illness ED visit rate (3-year baseline)	CDC PLACES; CDC WONDER; NSSP
Environmental / Meteorological	Land surface temperature (LST), urban heat island intensity, impervious surface	Landsat 8/9 via Google Earth Engine; Copernicus ERA5

	coverage, NOAA HeatRisk forecast level, maximum temperature anomaly, humidity anomaly, event duration (days), nighttime low temperature	reanalysis; NOAA CDO; NOAA HeatRisk API

Output: What the Model Produces

For each approaching heat event (defined as NOAA HeatRisk Level 3 or higher forecast, ≥ 48 hours out), SustAInable outputs:

- A probability score (0.00–1.00) per ZCTA representing predicted likelihood of elevated heat illness
- A ranked priority list of ZCTAs by risk score for city and CBO resource deployment
- A risk tier classification: High (score ≥ 0.60), Elevated (0.30–0.59), Baseline (<0.30)
- A feature contribution summary per ZCTA (SHAP values) to explain why a ZIP code was flagged

The SHAP explanations are a deliberate equity safeguard: they allow community organizations and residents to see the reasoning behind a risk flag, and to identify if a neighborhood is being scored based on factors they dispute or that reflect historical data gaps rather than current conditions.

Algorithm: XGBoost

SustAInable uses XGBoost (eXtreme Gradient Boosting), a gradient boosted decision tree ensemble. It is selected for the following reasons:

- Handles tabular data with mixed feature types (continuous, categorical, sparse indicators) without extensive preprocessing
- Tolerates missing values natively — critical given uneven coverage in state-level ED surveillance data
- Produces calibrated probability outputs, not just binary classifications
- Supports class weighting and threshold adjustment for the asymmetric false negative cost described below
- Benchmark precedent: An XGBoost model for heat illness prediction in South Korean cities achieved AUC = 0.895 in published peer-reviewed research (Kim et al., 2021). That is SustAInable's performance target.

Training and Holdout Design

Split	Years	Rationale

Training Set	2010–2021	12 years of heat events across US ZCTAs, providing exposure to regional variation and multiple heat event profiles. Excludes post-COVID data disruption.
Holdout Set	2022–2024	Intentionally includes the 2022 Pacific Northwest dome, 2023 Texas record heat, and 2024 Phoenix multi-week event as stress tests. These extreme events post-date the training window, testing whether the model generalizes beyond its training distribution.

The holdout set is treated as truly unseen data. No hyperparameter tuning is performed on holdout results. Final evaluation metrics are reported on the holdout set only.

Primary evaluation metrics:

- AUC-ROC: Overall discriminative performance (target: ≥ 0.85)
- Recall (sensitivity): Proportion of true elevated-illness ZCTAs correctly identified (target: ≥ 0.80, prioritized over precision)
- Precision: Of ZCTAs flagged as high-risk, proportion that actually experienced elevated illness
- Precision-Recall AUC: More appropriate than ROC for imbalanced classes (most ZCTAs will NOT have elevated illness during any given event)

The Two Ways the Model Can Be Wrong — and Why They Are Not Equal

Every prediction model makes two kinds of mistakes. SustAInable's design is shaped by a deliberate asymmetric cost judgment about these two error types.

Error Type	What Happens	Cost
False Positive	We flag a ZIP code as high-risk. The heat event arrives. The neighborhood does not experience elevated illness.	Cooling resources, outreach staff, and wellness check teams are deployed to a neighborhood that did not need priority attention during this event. Estimated cost: \$2,000–\$8,000 per ZCTA in misallocated resources. Recoverable. No one is harmed.
False Negative	We do NOT flag a ZIP code. The heat event arrives. The	Disabled people, elderly residents, and chronically ill

	neighborhood experiences elevated illness.	individuals in that neighborhood suffer preventable heat-related illness. Hospitalizations cost $15,000–$30,000 per admission. Deaths are irreversible. The harm is not recoverable.

The gap between these two outcomes is not close. A false negative is categorically worse than a false positive. This asymmetry drives every technical decision in SustAInable.

Specific measures to reduce false negatives:

- Lower decision threshold: ZCTAs are flagged as elevated-risk when the model probability score reaches 0.30, not the default 0.50. This catches more true elevated-illness events at the cost of more false alarms — a trade we accept.
- Class weighting: During training, the model is penalized more heavily for missing an elevated-illness ZCTA than for issuing a false alarm. XGBoost's scale_pos_weight parameter is set to reflect the inverse ratio of class frequency.
- SMOTE oversampling: Synthetic Minority Oversampling Technique is applied to the training set to increase the representation of elevated-illness events, improving the model's ability to learn the minority class pattern.
- Recall-prioritized threshold selection: Final threshold is selected by maximizing recall subject to a minimum precision floor of 0.40. We accept up to 2.5 false positives per true positive to ensure coverage.

Risks and Mitigations

Risk	Description	Mitigation
Label Construction Gap	CDC HHI data is a 3-year aggregate, not event-level. True event-year labels must be reconstructed from NSSP syndromic surveillance and WONDER data, which have uneven state coverage.	Begin with states with strong NSSP coverage (CA, TX, NY, FL). Document geographic limitations explicitly. Expand as coverage improves. This is a known constraint, not a hidden one.
Ecological Fallacy	ZIP-level predictions can obscure within-ZIP disparities. A wealthy ZIP with a low-income enclave may show a low average score while masking concentrated risk.	Layer SHAP feature contributions to show which factors drove a low score. Partner with CBOs who have block-level community knowledge to supplement ZIP-level outputs. Future iteration: move to census tract level.

Model Drift	Training data reflects 2010–2021 heat patterns. Climate change is shifting baseline temperatures and event profiles faster than historical data can capture.	Annual retraining cycle with rolling holdout evaluation. Monitor AUC degradation as leading indicator of drift. Weight recent years more heavily in training sample.
Equity Risk: Missing the Most Vulnerable	If training data underrepresents communities with the least access to healthcare (who may under-report heat illness in ED data), the model learns to underestimate their risk.	Perform stratified error analysis by income quintile and disability prevalence. Flag ZCTAs where model confidence is low due to sparse historical data. Treat data sparsity as a risk signal, not a clean bill of health.
Misuse as Sole Decision Tool	A probability score could lead a city official to under-respond to a ZCTA that scored 0.28 when the threshold is 0.30 — a false precision error.	SustAInable is a targeting aid, not an authority. All outputs are accompanied by confidence intervals and feature explanations. Deployment guidance explicitly states that scores inform, not replace, human judgment.

Stakeholders and Use Cases

SustAInable is designed for multi-stakeholder use. The same model produces outputs that serve different workflows at different levels of government and community:

Stakeholder	Use Case
Municipal Emergency Management (OEM)	Allocate cooling buses, open cooling centers in highest-risk ZCTAs 48–72 hours before event peak
Public Health Departments	Target door-to-door wellness checks and heat line outreach; prioritize elderly and Disabled resident registries for proactive calls
Community-Based Organizations (CBOs)	Deploy water and supply distribution to high-risk corridors; identify which blocks are inside vs. outside officially-flagged ZCTAs to supplement city response
Hospitals and Health Systems	Pre-position emergency capacity and staff in advance of predicted high-volume heat illness days
Disability Service Organizations	Flag ventilator-dependent and other medically fragile clients in high-risk ZCTAs for prioritized check-ins and power outage contingency planning

Performance Benchmark

SustAInable's performance target is grounded in published peer-reviewed research:

- Kim et al. (2021), International Journal of Environmental Research and Public Health: An XGBoost model for predicting heat illness risk at the neighborhood level in South Korean cities achieved AUC = 0.895 on holdout data. SustAInable adopts this as its minimum performance target.
- Recall target of ≥ 0.80 is set independently of AUC, reflecting the asymmetric cost judgment described above. A model with AUC = 0.89 but recall = 0.65 is not acceptable for SustAInable's use case.
- Baseline comparison: A structural vulnerability index alone (CDC HHI decile ranking) produces an AUC of approximately 0.72–0.74 in published evaluations. SustAInable must meaningfully outperform this baseline to justify development.

Data Sources

All datasets listed below are publicly accessible without purchase or proprietary agreement:

Dataset	Use	Provider
CDC Heat and Health Index (HHI)	Structural vulnerability features at ZCTA level	CDC Climate and Health Program
ACS 5-Year Estimates	Poverty, housing age, disability prevalence, renter rate, household density	US Census Bureau
CDC PLACES	Chronic disease prevalence by ZCTA (CVD, COPD, obesity, diabetes)	CDC Division of Population Health
NOAA Climate Data Online (CDO)	Historical temperature, humidity, heat event records	NOAA National Centers for Environmental Information
NOAA HeatRisk	Forecast heat danger level (0–4) for training event matching	NOAA Weather Prediction Center
Landsat 8/9 Surface Temperature	Land surface temperature and urban heat island intensity by ZCTA	USGS / Google Earth Engine
NLCD 2021	Impervious surface coverage and land cover classification	Multi-Resolution Land Characteristics Consortium
Copernicus ERA5 Reanalysis	Historical temperature anomaly and humidity at ZCTA level	ECMWF / Copernicus Climate Change Service
NSSP (National Syndromic Surveillance Program)	Emergency department visit records for heat illness label construction	CDC Center for Surveillance, Epidemiology & Lab Services
CDC WONDER	Heat-related cause of death records for label validation	CDC National Center for Health Statistics

Closing: What Building This Tool Means

Heat-related deaths have risen 117% since 1999. The communities most at risk are consistently the ones with the least capacity to adapt — the ones where structural disinvestment, inadequate housing, and the highest concentration of Disabled, elderly, and low-income residents compound into a predictable pattern of preventable harm.

The tools to get ahead of that pattern exist. The data is publicly available. The algorithmic methods are proven in analogous international contexts. The gap is not technical capacity. The gap is whether we choose to build for the communities most at risk — before the crisis, not after.

SustAInable is a proposal to close that gap, one heat event and one ZIP code at a time.

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Key Citations

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